

24.1 A man stands 1m from a vertical plane mirror. What is the distance from the man to his image?

24.4 An eyeglass lens has one concave surface of radius 0.5m and convex surface with radius 0.7m. If the of refraction of the lens is 1.6, find the focal length.

24.10 A lens has a focal length of 0.2m. A real object is placed 0.08m from the lens. (a) Locate the image graphically (b) Locate the image algebraically (c) What is the magnification?

24.11 An object is placed 1m from lens with a focal length of -0.5m. Locate the image position (a) graphically (b) algebraically.

24.18 A doctor finds that a nearsighted person needs one lens with power of -8 diopters and one with power of -6 diopters. Find the focal lengths of the lenses.

24.19 Find the focal length of lens with a power of 4 diopters.

24.53 A concave shaving mirror with a radius of 0.3m is held so the image man is three times as large as his face. How far is the mirror from his face?